11 The first three terms of a geometric sequence are $5k-2$, $3k-6$, $k+2$, where k is a constant.

(a) Show that k satisfies the equation $k^2-11k+10=0$. **[3]**

(b) When k takes the smaller of the two possible values, find the sum of the first 20 terms of the sequence. **[3]**

(c) When k takes the larger of the two possible values, find the sum to infinity of the sequence. **[2]**